Darsh Butala

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Aerospace Engineering postgraduate at IIT Bombay with expertise in high-performance propulsion system design and control system development. Experienced in experimental combustion (DDT & Hybrid Rockets), CFD-based optimization, dynamic modelling, and flight stability analysis using MATLAB/Simulink. Seeking opportunities across space, defence, aero-engine, or UAV applications involving propulsion, flight dynamics.

SKILLS

Simulation & Engineering Analysis

- ANSYS Fluent, ANSYS Mechanical
- MATLAB & Simulink

CAD & System Integration

- SolidWorks, CATIA, AutoCAD,

Programming

- Python, Java, C / C++

INTERNSHIPS & PROJECTS

IIT Bombay | M. Tech Thesis | *Jul 2025 – May 2026*

LPG-Fuelled Deflagration-to-Detonation Transition (DDT) for High-Intensity Acoustic Systems

- Designed and fabricated a compact LPG–air DDT test section with controlled ignition and purge cycles ensuring safe and repeatable operation.
- Conducted combustion experiments capturing high-frequency pressure-time data to characterize detonation wave structure and transition behaviour.
- Analysed shock dynamics and acoustic impulse generation for propulsion-based field applications.
- Integrated combustion theory, gas dynamics, and experimental diagnostics for system-level validation.

IIT Bombay | M. Tech Research Work | *Jan 2025 – May 2025*

Hybrid Rocket Regression Study

- Conducted comparative analysis of GOX and N_2O oxidizers for HTPB-based hybrid propulsion systems.
- Evaluated regression rate limitations and investigated nano-aluminium additives for ~25% potential regression enhancement.
- Assessed performance trade-offs in hybrid motor design through literature synthesis and analytical modelling.

B. Tech Project | Thrust Vector Control (TVC) Model Rocket | *Jan 2024 – May 2024*

- Designed and simulated a servo-actuated thrust vector control mechanism for trajectory correction.
- Performed stability and flight dynamics analysis using MATLAB and Open Rocket.
- Fabricated and validated a working prototype demonstrating controlled thrust vectoring.

Design & Analysis of 100 kN LOX–Methane Rocket Propulsion System

- Performed chemical equilibrium and performance analysis using NASA CEA, estimating chamber conditions, thrust, and specific impulse.
- Generated optimized nozzle contours using Rocket Propulsion Analysis (RPA) for targeted expansion ratios and operating conditions.
- Conducted CFD simulations in ANSYS Fluent to evaluate flow expansion and performance efficiency.
- Designed cryogenic propellant tanks in SolidWorks and performed structural validation in ANSYS Mechanical under 12 atm internal pressure.
- Integrated engine, tank, and structural subsystems into a modular propulsion stage architecture.

Design and Control of Quadcopter Drone

- Developed mathematical models of quadcopter dynamics in MATLAB/Simulink and implemented a PID-based control architecture for stabilization.
- Designed a lightweight frame in SolidWorks, validating stability and response characteristics through simulation.

STAR Lab, Surat | Rocket Testing & Avionics Intern | *Jun 2022 – Jul 2022*

Static Test Pad Development

- Designed and implemented a static test platform for solid and hybrid rocket motors.
- Integrated load cell-based thrust measurement system for real-time performance monitoring.
- Established safety mechanisms including remote ignition, blast shielding, and emergency shutdown protocols.

High-Powered Rocket Avionics

- Developed onboard avionics integrating GPS, telemetry, and recovery systems.
- Programmed Arduino-based data logging for altitude, acceleration, and orientation tracking.

EDUCATION

M. Tech, Aerospace Engineering (Specialization: Propulsion) — **Mumbai, INDIA**
Indian Institute of Technology Bombay | IIT B — 2024 – 2026
CPI: 8.23

B. Tech, Aerospace Engineering — **Mumbai, INDIA**
Amity University Mumbai — 2020 – 2024
CGPA: 8.78

Scholastic Achievement:
Secured All India Rank 106 in GATE 2024 (Aerospace Engineering)

ADDITIONAL INFORMATION

Certifications

- Simulation using ANSYS Fluent – Udemy
- ANSYS Workbench – Udemy
- Learning SOLIDWORKS – CADCIM Technologies
- MATLAB & Simulink Onramp – MathWorks
- Space Entrepreneurship 101 – Udemy
- Introduction to Tech Entrepreneurship – Coursera

Positions of Responsibility

- Teaching Assistant – Gas Dynamics, IIT Bombay (Jul 2025 – Dec 2025)
  Guided 30+ students through assignments, evaluations, and technical doubt resolution.